



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Antweight Battle Bot

Electrical & Computer Engineering

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04/30/2026

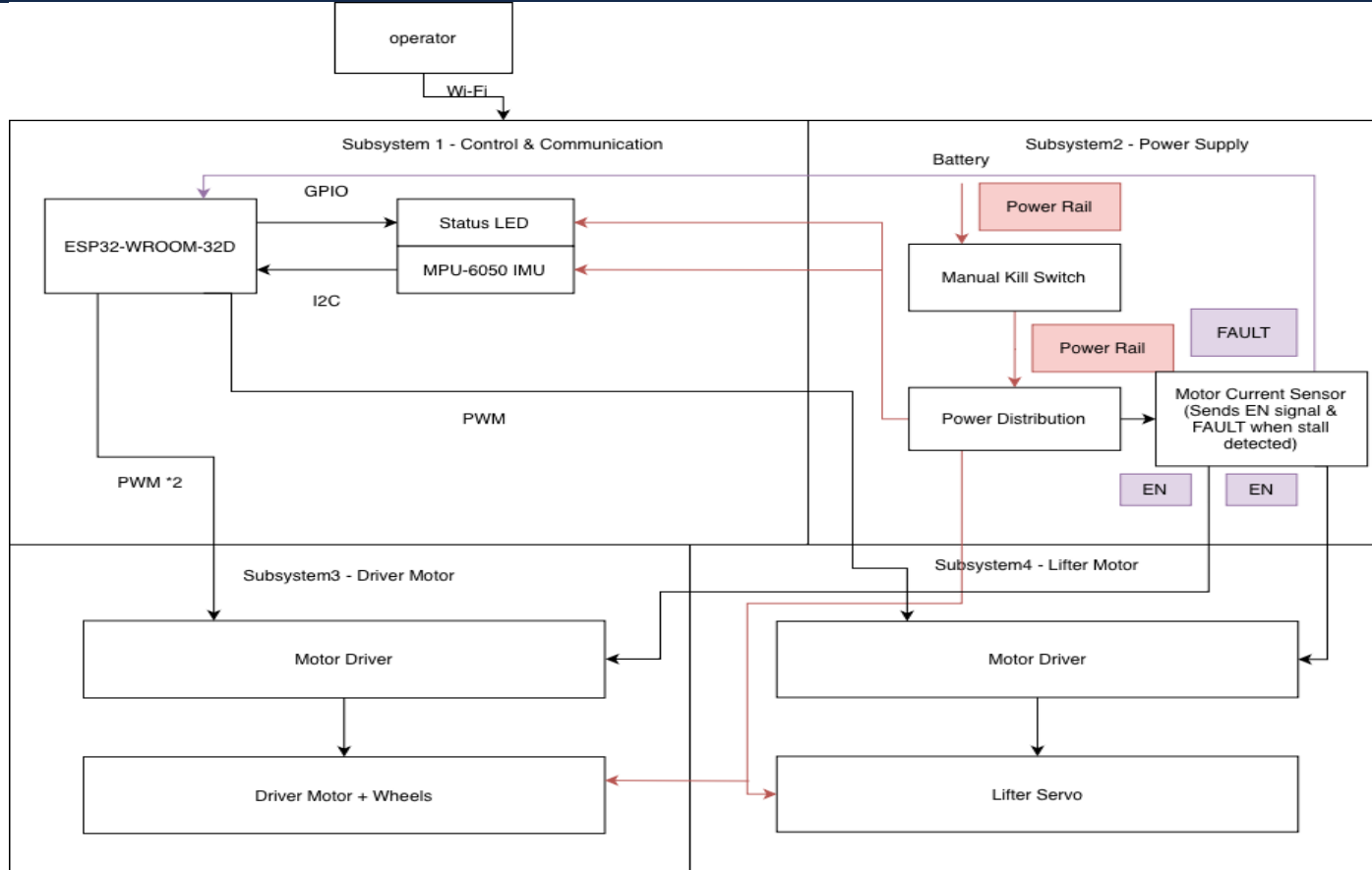
Objectives/Constraints

- Build a safe, controllable ≤ 2 lb antweight battlebot
- Survive impacts while maintaining mobility and control
- Operate reliably during a 2-minute match
- Avoid brownouts during stalls and heavy motor loads

High Level Requirements

- Weapon: lift 0.9 kg $\geq$ 40 mm within 2 s; hold $\geq$ 5 s
- Control: wireless PC control with latency $\leq$ 150 ms
- Safety: kill switch stop $\leq$ 0.5 s; link-loss stop $\leq$ 2 s
- Mobility: speed $\geq$ 0.4 m/s; push 1.0 kg for $\geq$ 2 s without brownout

Block Diagram

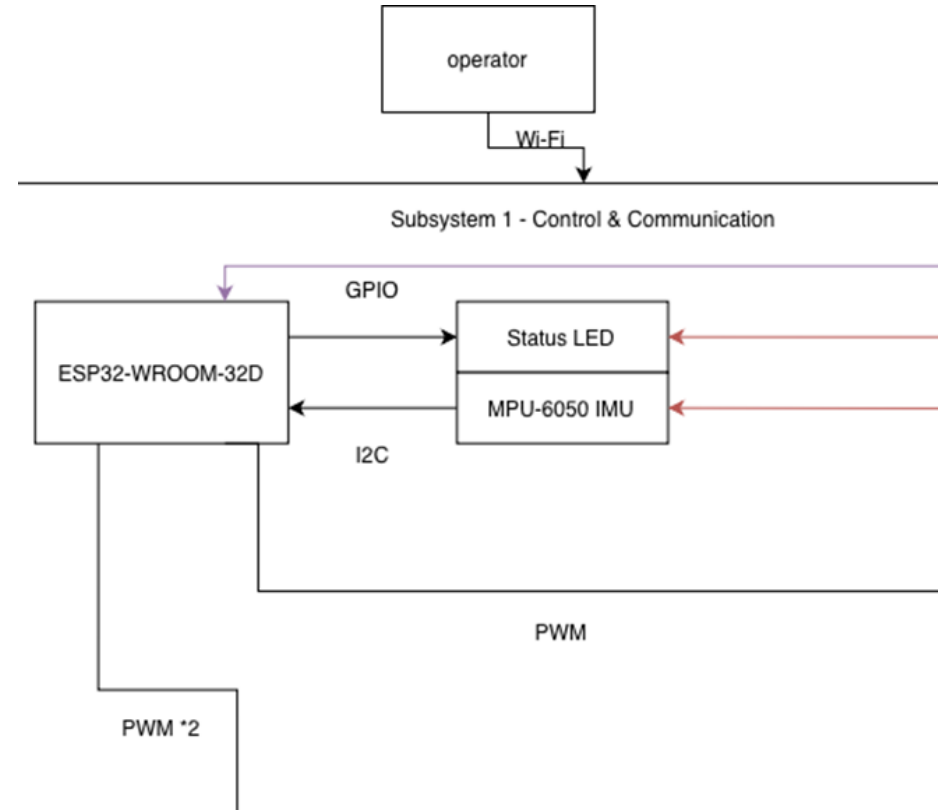


Control & Communication Subsystem

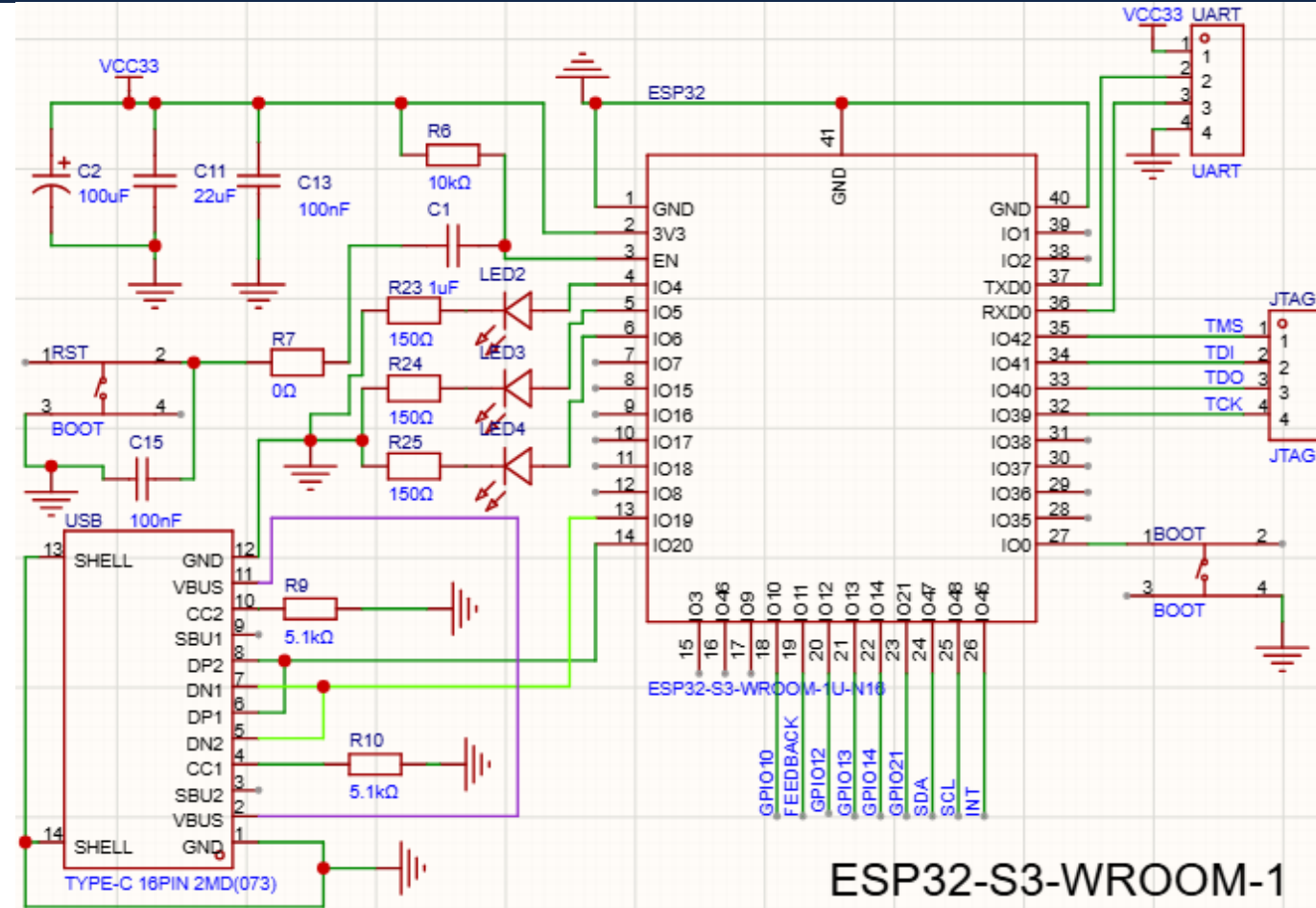
Subsystem Overview



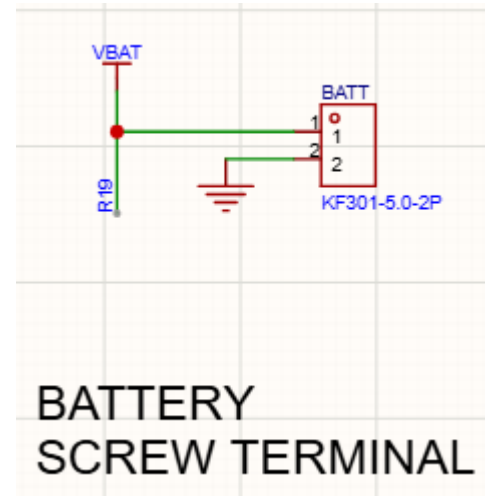
- WiFi commands
- ESP32-S3 sends signals
- Codes handle additional features



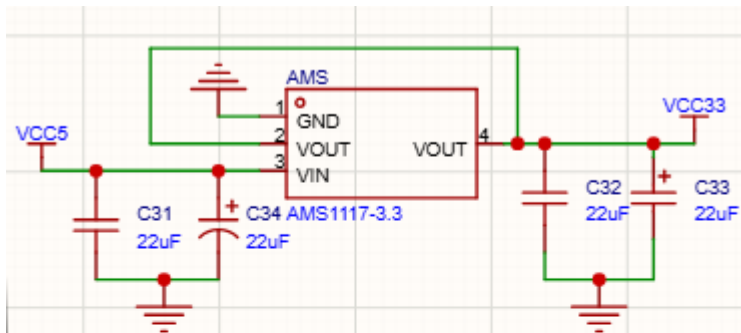
Hardware Schematic



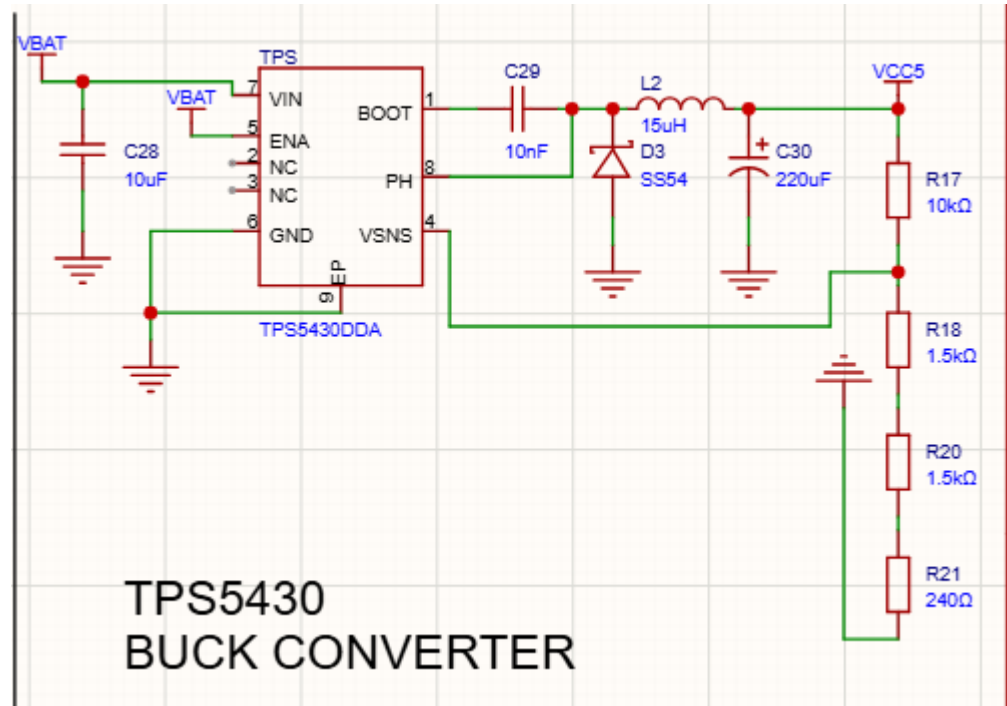
ESP32-S3-WROOM-1



Hardware Schematic

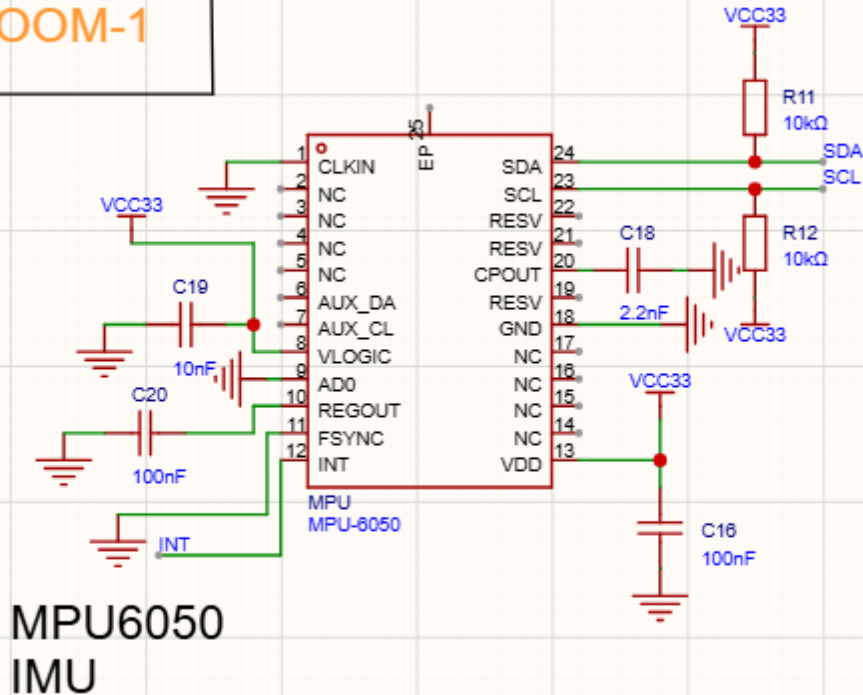


ASM1117-3.3
VOLTAGE REGULATOR

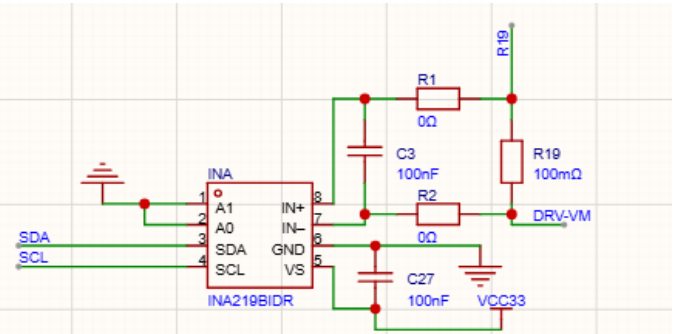


TPS5430
BUCK CONVERTER

WROOM-1

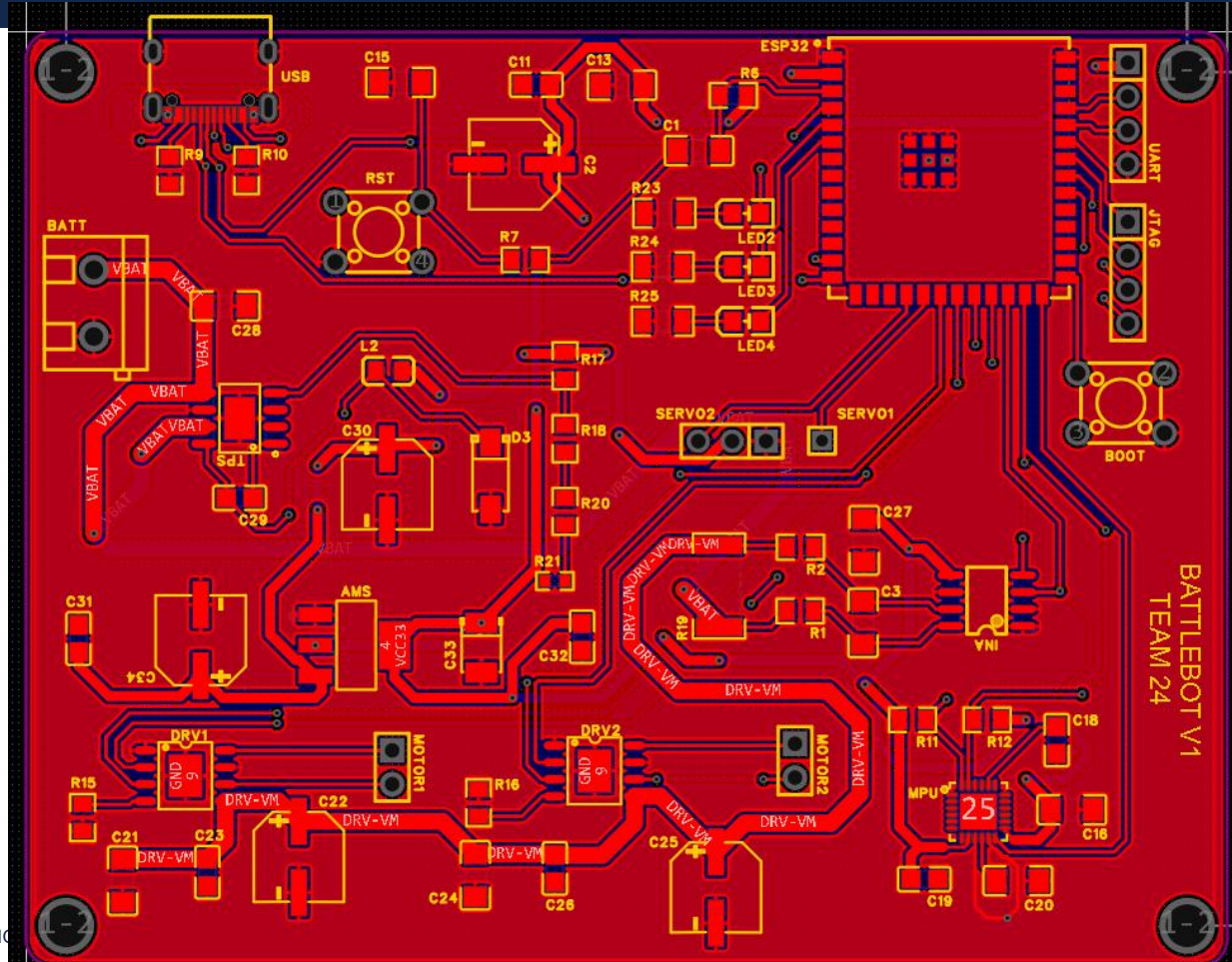


MPU6050
IMU

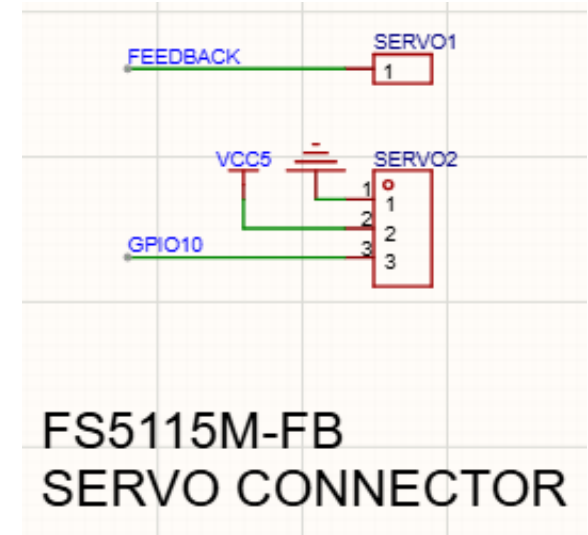
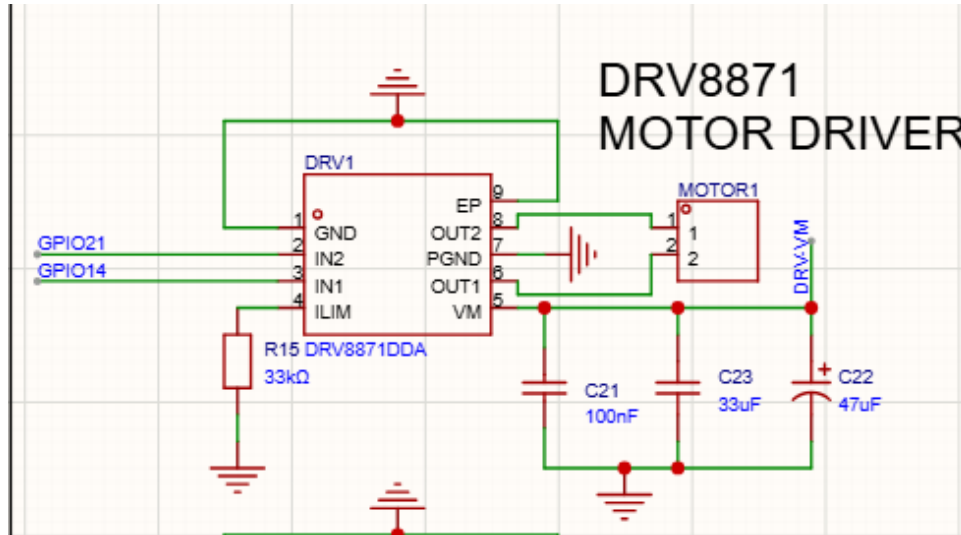


INA219
CURRENT MONITOR

PCB Layout



Websocket → ESP32 → PWM(Pulse Width Modulation) → motor/servo signal



- ESP32 as embedded WebSocket server using softAP
- Transmission through TCP
- Networking through IP
- Data through Wifi

Software Features



OBJ 279 // CONTROL CONSOLE

UPTIME 00:00:13

CONNECTED

DRIVE CONTROL

JOYSTICK + SPEED



LIFTER SERVO

Q / E

▲ UP

▼ DOWN

POS --

KEYBOARD

W A S D drive (combine for diagonal)
Q lifter up E lifter down
WA=fwd-left WD=fwd-right SA=rev-left
SD=rev-right

IMU VISUALIZATION

MPU6050

HEADING 2D



YAW
60.1°

CMD
--

ATTITUDE 3D



ROLL
-31.0°

PITCH
-33.9°

YAW
60.1°

RAW IMU

ACCEL G · GYRO °/S

	X	Y	Z
ACCEL	-0.01	0.00	0.99
GYRO	-0.09	-0.05	0.00

Sensor Temperature

27.4 °C

POWER MONITOR

INA219

CURRENT

-1.50 mA

BUS VOLTAGE

11.124 V

POWER

16.00 mW

LOAD VOLTAGE

11.124 V

SHUNT

-0.150 mV

CURRENT - LAST 30S

-1.5 MA

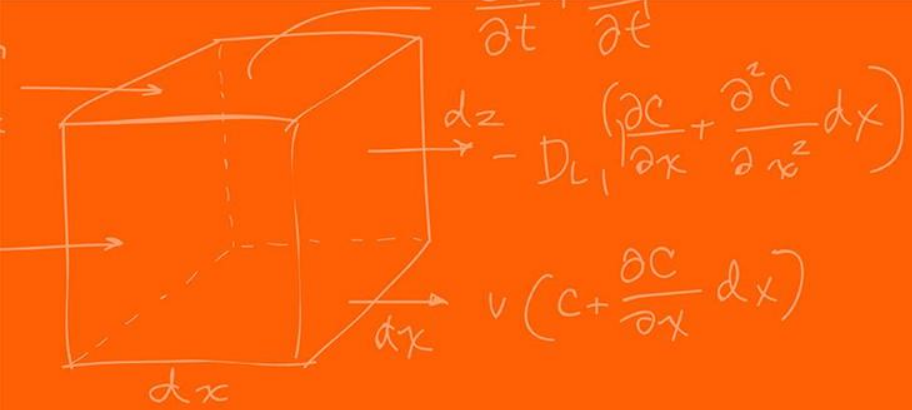


BUS VOLTAGE - LAST 30S

11.12 V



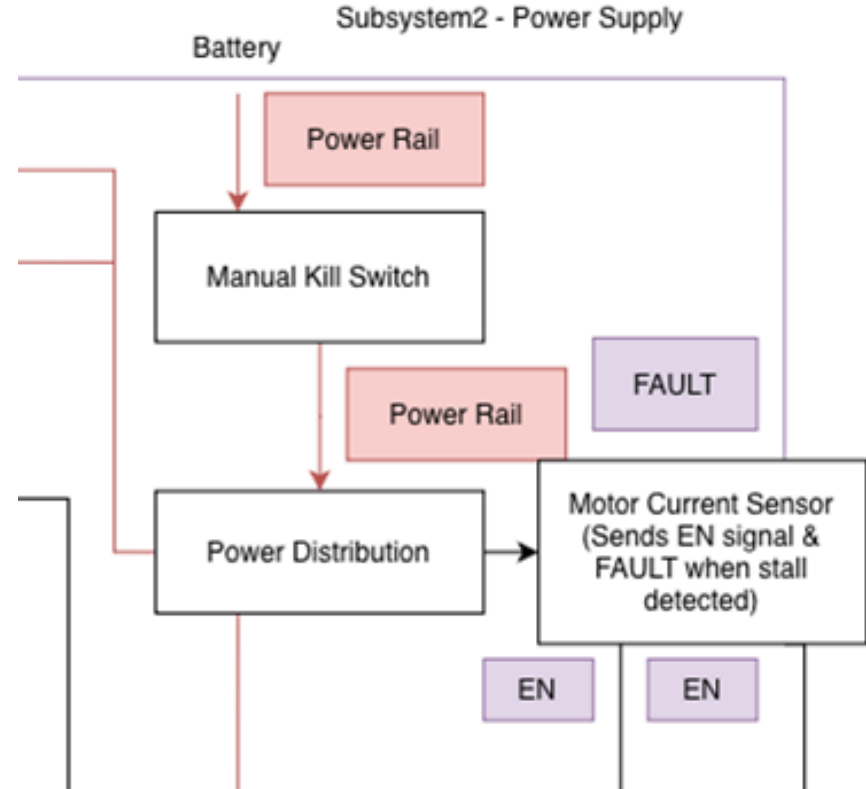
- ✓ 3.3 V rail: 3.36 V avg, within $\pm 5\%$
- ✓ Safety: link-loss/fault sets PWM = 0%



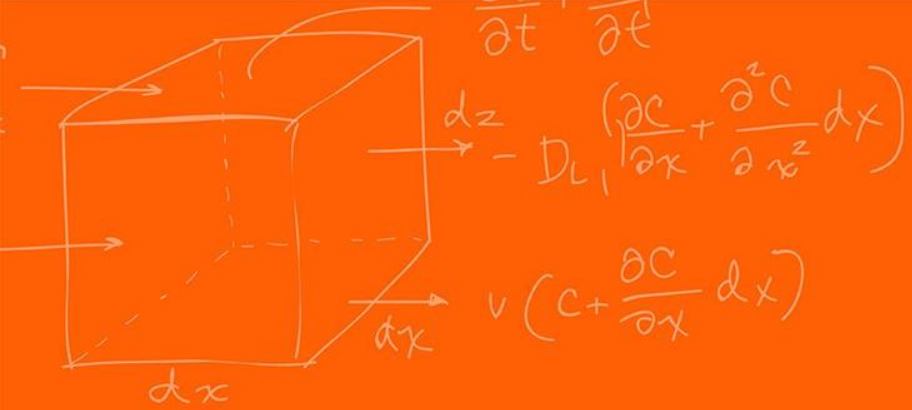
Power Supply & Safety Subsystem



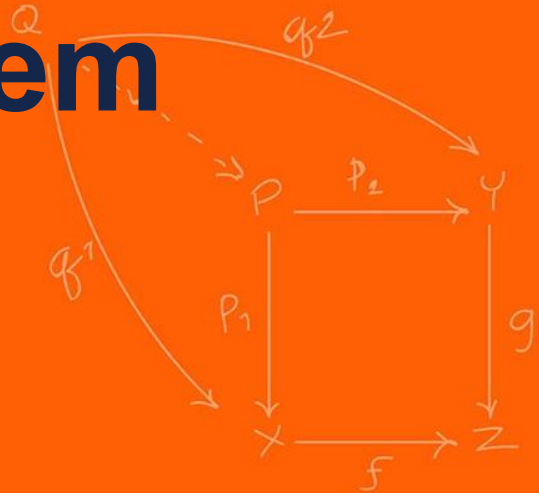
- Battery supplies motor and major power rail
- 3.3 V powers ESP32 sensor chips
- Kill switch and fault protection disable motion safely



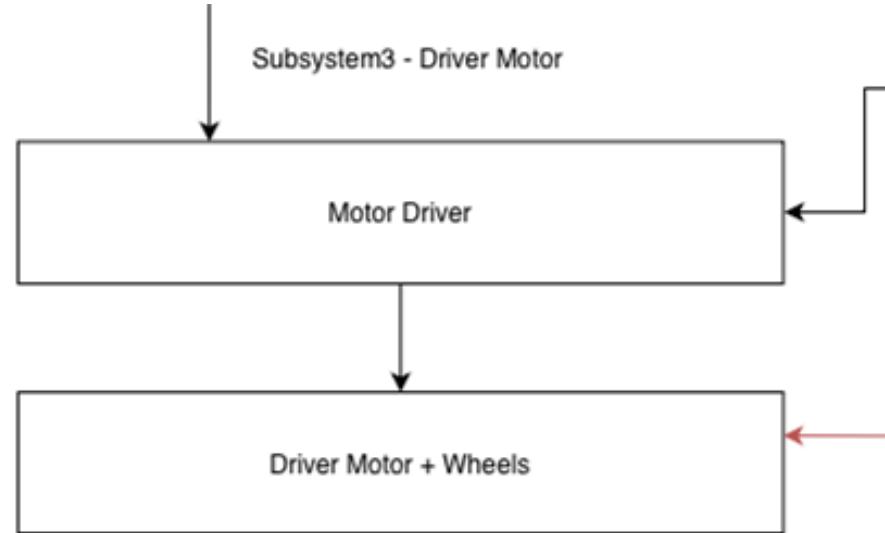
- ✓ Buck output: 3.24 V avg
- ✓ Ripple: 60 mV
- ✓ Error: 1.8% → within $\pm 5\%$
- ✓ Safety: kill switch cuts motor power



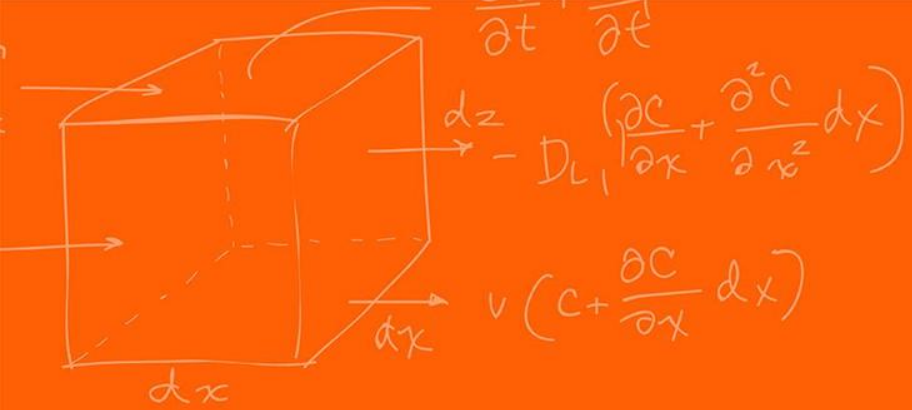
Drive Subsystem



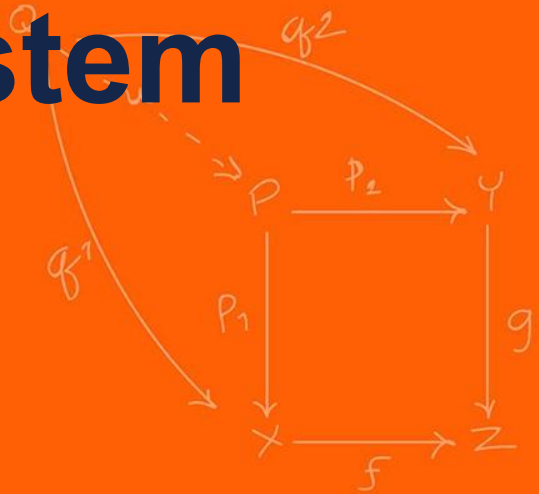
- DRV8871 drivers control motors
- Supports speed limiting



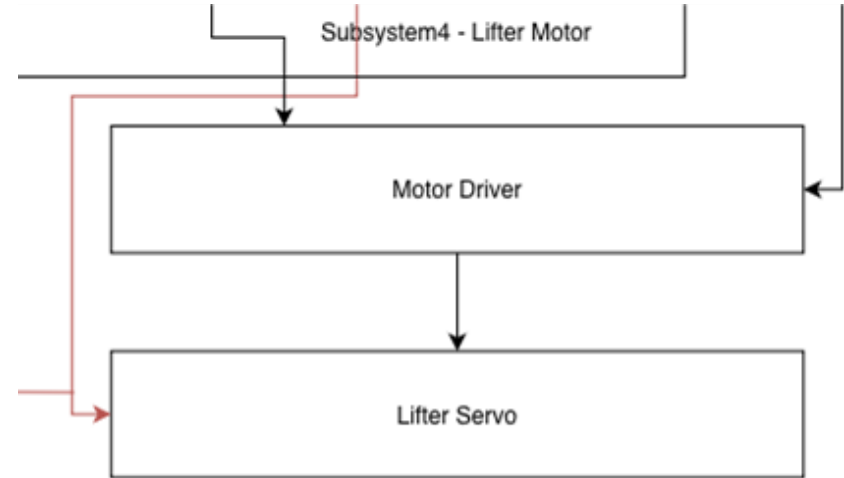
- ✓ Drive speed: measured 1310 RPM at 25% throttle
- ✓ Expected speed: 1562.5 RPM
- ✓ Speed error: 16.6%
- ✓ Cause: friction losses from tangential wheel drive



Weapon Subsystem



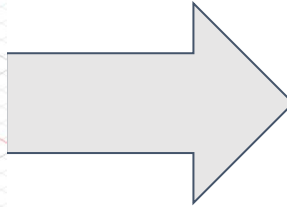
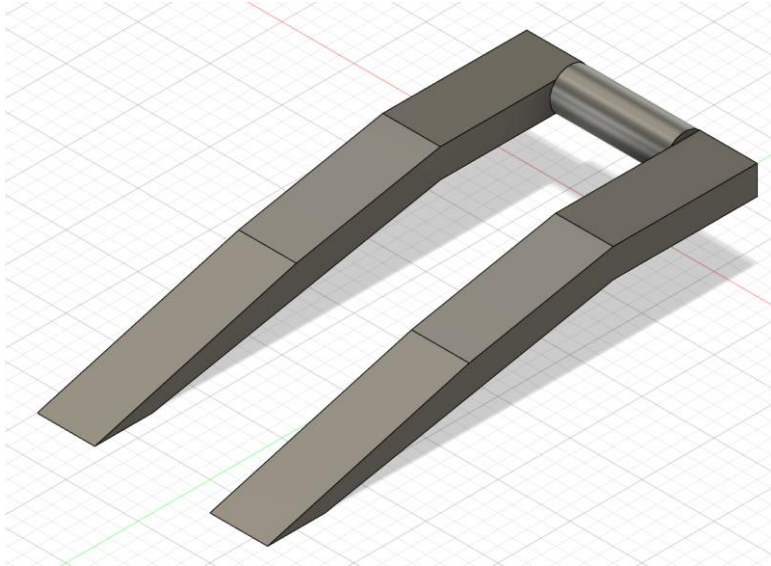
- Servo receives direct GPIO signal
- Large torque servo ensures force output



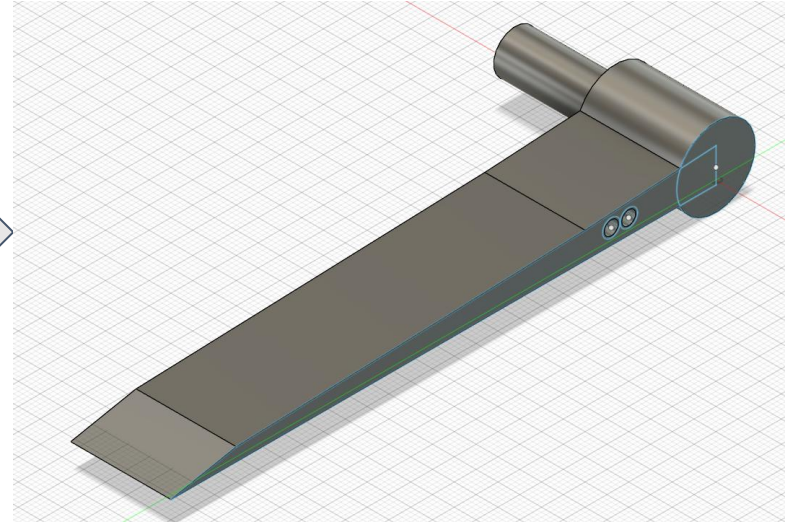
Weapon Motor Changes

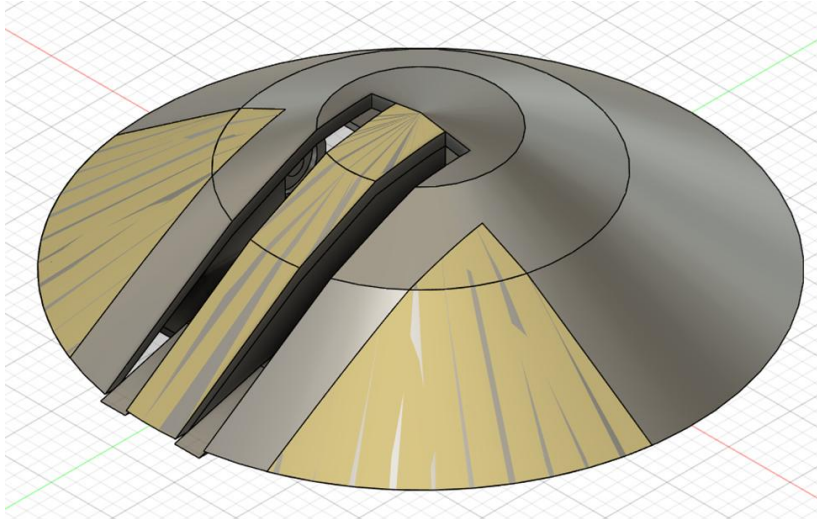


Original Design

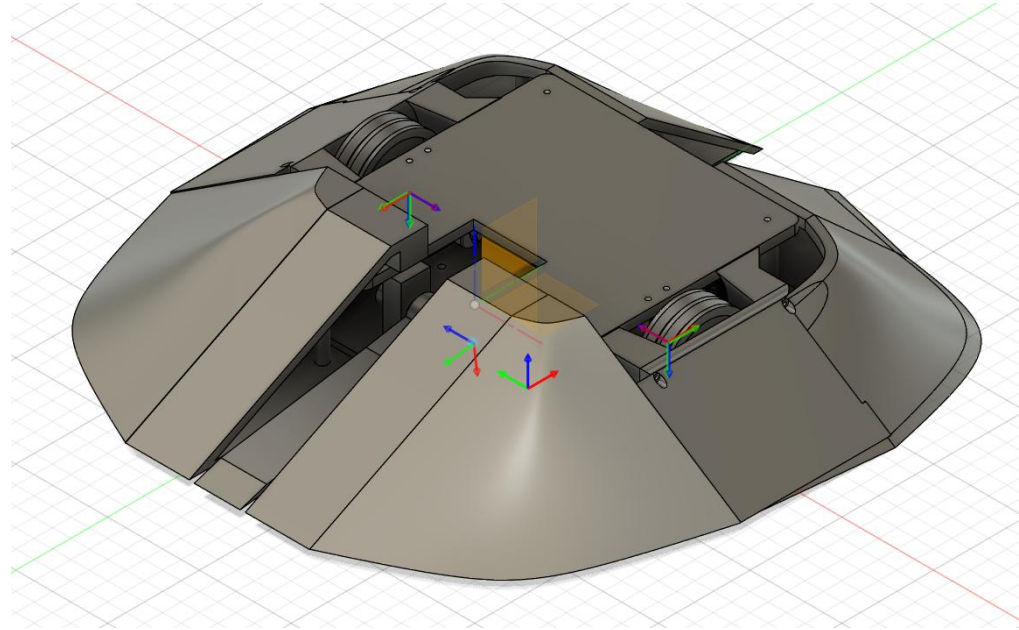


New Design



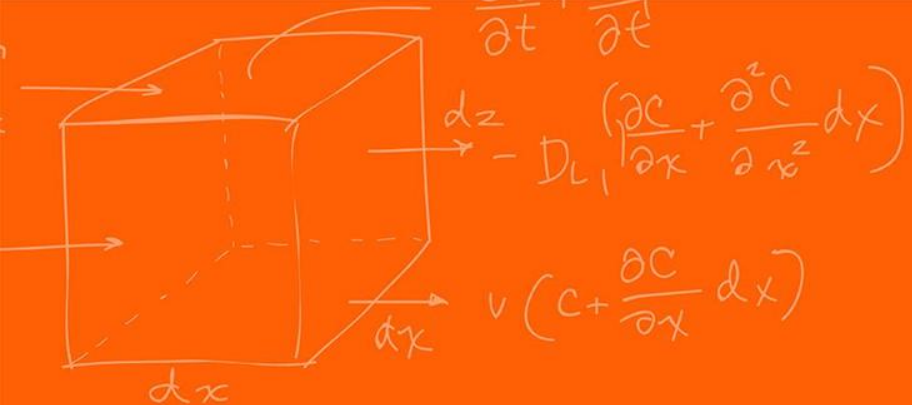


Version 1

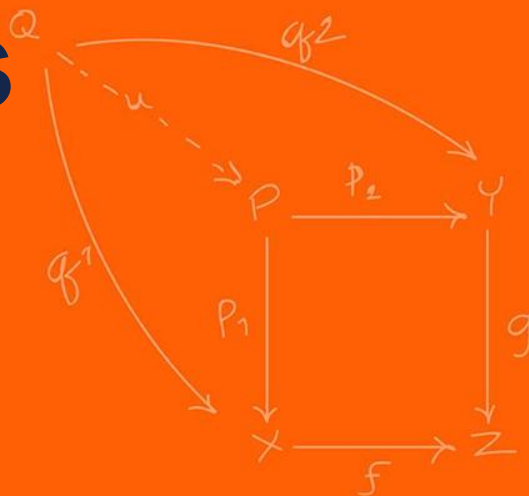


Version 2

- ✓ Throttle limit: tested up to 22.5%
- ✓ Tip speed check: surpassed 150 mph target
- ✓ Calculation: $v = 2\pi r \times \text{RPM} / 60$
- ✓ Design change: larger weapon radius increased tip speed



Discussions



Successes

- Wifi control successfully sends motor commands to ESP32-S3
- Mechanical redesign improved chasis stability

Challenges

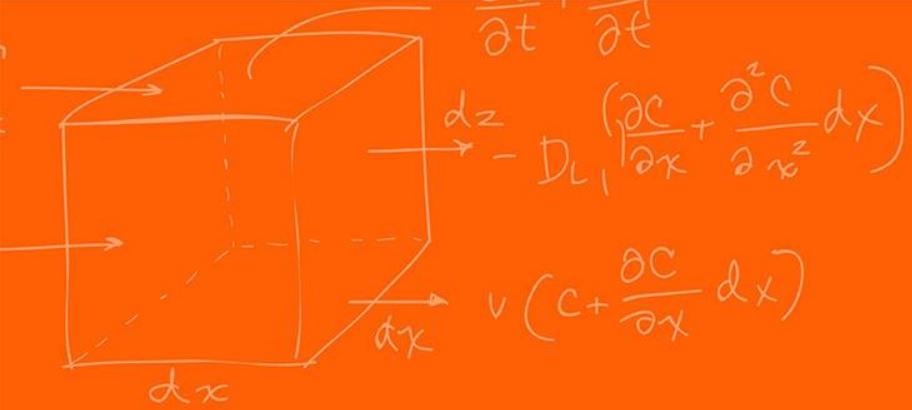
- PCB bring-up required careful I2C/PWM initialization
- Integration space was limited inside the compact chassis

What We Learned

- Real measurements are critical for validating design assumptions
- Always start early
- Mechanical packaging strongly affects electrical and software choices

What Are The Next Steps

- Improve chassis and weapon contact surfaces
- Refine software UI for easier driving and debugging



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